

Alpilla 600 MW Combined Cycle Power Plant

Engineering execution plan (workflow, interfaces, timeline)

Document number	ALP-EPC-00000-GE-PLN-0001
Revision	draft - not yet issued (next issue: IFR rev A)
Originator	Istanbul EPC
Review class	approval
Planned issue (workflow timeline)	IFR 2027-03-03, IFA 2027-03-31, IFC 2027-04-07
Basis	requirement:ER-01.09

1. Purpose and scope

This plan sets how the engineering of the project is executed: scope (the MDL), organisation, the workflow between documents and disciplines, the review cycle and the timeline against the construction need dates (Advanced Work Packaging, ALP-EPC-00000-GE-PLN-0002). All figures are taken from the MDL and the workflow timeline computed from it; they change when the records change.

2. Organisation and responsibilities

Party	Responsibility
Istanbul EPC	Consortium leader: plant design and balance of plant engineering, integration of the power island, MDL and document control, procurement of BOP packages, construction and commissioning management
Imaginary Electric	Power island design and supply (GT, HRSG, ST, generators, GSUs, TCS); VDRL documents per the MDL dates (TQ-IEC-023); interface data (IF- points)
Owner	Review and approval per the review class (14 / 21 days); approval of the KKS key list, numbering and AWP (TQ-035)
Engineer	Owner's Engineer: technical review on behalf of the Owner, KKS key verification

3. Engineering scope (MDL)

Discipline	Docs	EPC	Supplier	BDP	In EWP	Hours	First IFR	Last IFC
civil	141	115	26	1	73	29,888	2026-11-02	2027-09-16
commissioning	212	206	6	0	0	7,430	2026-11-23	2028-05-31
electrical	184	137	47	2	112	16,506	2026-09-28	2027-12-13
general	181	108	73	2	0	6,112	2026-09-28	2027-12-28
hse	38	37	1	1	11	3,924	2026-12-07	2027-10-19
hvac	22	22	0	0	14	4,850	2026-11-02	2027-07-12
i and c	425	333	92	2	58	28,480	2026-11-02	2028-01-26
mechanical	656	196	460	1	42	13,418	2026-09-28	2028-07-24
pipng	152	150	2	3	112	74,788	2026-11-02	2028-07-06
process	153	126	27	13	0	18,007	2026-09-28	2027-08-13
project control	2	2	0	0	0	160	2027-02-24	2027-05-12
structural	42	26	16	0	37	8,282	2026-11-02	2027-07-28

Total 2208 documents, 211,845 hours: EPC document preparation incl. checking and the plant 3D model; for supplier documents the EPC review. Site support, expediting and management are budgeted separately.

3.1 How the MDL is derived

80 MDL rules (records mdl_rule) define the documents per plant, system, structure, construction area, procurement package and equipment group: type, quantity basis (sheets per P&ID sheet, instrument, loop, line, isometric, consumer, structure ...), hours, inputs, EWP / MR links and, for supplier documents, the timing after the purchase order. Quantities are system estimates (basis DEC-EPCE-0008) until the equipment, instrument and line registers are larger; python -m engine mdl sync keeps the documents in line with the rules and registers. Single deliverables (studies, plans, permit dossiers, test reports) are individual records citing their requirement, permit or clause.

Rule scope	Rules
cwa	10
eqgroup	3
mr	24
plant	11
structure	14

Rule scope	Rules
system	18

3.2 Completeness against indicative benchmarks

Indicative ranges for a comparable 1x1 H-class CCGT (engineering judgement, not contractual): below the minimum = scope probably missing (WARN), above the maximum = review for duplication (INFO).

Benchmark	Scope	Measure	MDL	Indicative range
BM-ALL-EPC	all / EPC	sheets	13,382	8,000 - 20,000
BM-ALL-EPC	all / EPC	hours	201,300	150,000 - 320,000
BM-ALL-SUP	all / SUPPLIER	sheets	7,983	3,000 - 10,000
BM-CM	commissioning / EPC	docs	206	150 - 400
BM-CV	civil / EPC	sheets	704	500 - 1,500
BM-EL	electrical / EPC	sheets	1,876	1,200 - 3,000
BM-GE	general / EPC	docs	108	50 - 200
BM-HS	hse / EPC	docs	37	30 - 120
BM-HV	hvac / EPC	sheets	156	100 - 400
BM-IC	i and c / EPC	sheets	5,259	4,000 - 9,000
BM-ME	mechanical / EPC	sheets	469	250 - 800
BM-PI	pipng / EPC	sheets	4,073	3,500 - 7,000
BM-PR	process / EPC	sheets	248	200 - 500
BM-SS	structural / EPC	sheets	223	150 - 600

Coverage is checked at every engine run: every design discipline has its design criteria, every KKS system has the document types its category requires, every Employer's Requirement and every EPC / IEC scope item is answered by at least one document, every equipment item has a datasheet, an MR and a CWP, every MR a specification and a requisition, every CWP an EWP.

4. Workflow rules

- Documents form a network through their inputs (MDL column Inputs). A document starts when all its inputs have reached the maturity its type requires (IFR or IFC, section 4 of ALP-EPC-00000-GE-PRC-0002), not before NTP.
- First issue (IFR) = start + preparation time of the type (+5 % per extra sheet, at most double).
- Owner review 10 wd (15 wd for the Basic Design Package); approval class: IFA = IFR + review + update, IFC = IFA + 5 wd; review class: IFC = IFR + review + update; information / internal: IFC = IFR + update.
- Tender-stage documents (bid design, and the requisitions of long-lead packages per the procurement decision) are issued at NTP. Supplier documents with committed dates keep them (IEC, TQ-IEC-023); the others are timed from their MR: weeks after the PO for design data, weeks before delivery for FAT, erection and O&M documents. Issued revisions replace the planned dates.
- Construction documents belong to the EWP of their CWP. An EWP is needed ewp_lead_days before its CWP starts for the first IWPs; documents of progressive types (isometrics, supports, schematics, loop diagrams, reinforcement, equipment foundations ...) are released to later IWPs at CWP start - lead + share x CWP duration (share per document type, section 4 of ALP-EPC-00000-GE-PRC-0002). EPC documents that need certified vendor data (e.g. equipment foundations) take the supplier documents as inputs.
- Requisitions start the PWP (MR) cycle: PO = MRQ IFC + bid + award; delivery on site = PO + manufacture + transport; needed at the first CWP start (+ need lag for materials installed late in the CWP).
- Engineering capacity per discipline (eng_resource) with a mobilisation ramp from 30 % at NTP; documents are scheduled in need-date priority within the capacity (levelled timeline).

5. Interdisciplinary inputs (who feeds whom)

Number of input links from the supplying discipline (row) to the receiving discipline (column).

from \ to	CV	CM	EL	GE	HS	HV	IC	ME	PI	PR	PC	SS
CV	213		24	25	12	11	11		4			37
CM		198										
EL	4	12	236	12			10	77				
GE	1		5	76	3	2	12	71	3	43	3	
HS		1	1	1	45		41					1
HV	11	4	2	1		22	9	16	2	3		
IC		47	5	15			407	38		1		
ME	19	36	22	33	1	4	46	581	5	22		1
PI	41		26	9	8	4	11	15	357	41		15
PR	11	144	1		40	3	249	112	95	226		
PC												
SS	1			16				1				35

Typical inputs per document type (checked at every run; a document without any of them is reported):

Type	Title	Inputs at	Typical inputs
ARC	Architectural drawing	IFR	LAY
BLK	Block / architecture diagram	IFR	DBR, IOL, SLD
CAL	Calculation	IFR	CAL, DBR, HMB, LST, PID, RPT, SDS, SLD, STU
CBL	Cable schedule and routing	IFR	SLD, LAY
CEM	Cause and effect matrix	IFR	PID, CNR, HAZ
CMP	Commissioning procedure	IFR	CAL, CNR, PID, PLN, SPC, STU
CNR	Control narrative / functional description	IFR	PID, CNR
CVL	Civil drawing (grading, roads, drainage, marine)	IFR	PLT, STU, SDS
DSH	Datasheet	IFR	PID, SPC
EFD	Equipment foundation drawings (vendor-certified loads)	IFR	CAL, FND
ELY	Electrical layout (earthing, lighting, trays, equipment)	IFR	ARC, CVL, ELY, FND, LAY, PPL, SLD, STU
FND	Foundation / underground drawing	IFR	LAY, CAL
GAD	General arrangement drawing	IFR	ARC, DBR, LAY
HAZ	Safety study (HAZID, HAZOP, SIL, hazardous areas, fire, QRA)	IFR	PID, PLT, HAZ
HMB	Heat and mass balance	IFR	DBR
HMI	Operator display (HMI graphics) drawings	IFR	PID, CNR
HUP	Instrument hook-up drawings	IFR	PID, SPC
IOL	I/O list	IFR	PID, CNR, LST, IOL
ISO	Isometric drawing package	IFR	PPL
ITP	Inspection and test plan	IFR	SPC
LAY	Equipment layout / arrangement	IFR	ARC, HAZ, PLT
LOG	Logic diagram	IFR	CNR, IOL, RPT
LOP	Loop / connection diagram	IFR	IOL
LST	List / index (equipment, lines, valves, instruments, loads, tie-ins)	IFR	DBR, HAZ, LST, PFD, PID, PRC, RPT, SLD, STU
M3D	3D model review report (30 / 60 / 90 %)	IFR	PLT, LAY, PPL, ISO
MDB	Manufacturing record book / conformity certificates / turnover dossier	IFR	CMP, ITP
MOS	Method statement / construction execution plan	IFR	CVL, PLN
MRQ	Material requisition	IFR	SPC
OMM	Operation and maintenance manual	IFR	CMP, PID
PFD	Process flow diagram	IFR	HMB
PID	Piping and instrumentation diagram	IFR	SDS
PLN	Plan (execution, quality, HSE, commissioning, training)	IFR	DBR, HAZ, LST, PFD, PLN, PRC, RPT, SPC, STU
PLT	Plot plan	IFR	DBR
PPL	Piping general arrangement / plans	IFR	PID, LAY
RBR	Reinforcement drawings	IFR	FND
SCH	Schedule	IFR	MDL
SDS	System design description	IFR	DBR
SLD	Single line diagram	IFR	DBR, LST
STL	Steel structure drawing	IFR	GAD, CAL
STR	Pipe stress analysis report	IFR	PPL
SUP	Pipe support drawings	IFC	ISO, STR
TRP	Test report / test record / certificate	IFR	CMP, SPC, TST
TST	Test procedure (FAT, SAT, site and performance tests)	IFR	DBR, ITP, RPT, SPC
WIR	Schematic and wiring diagrams	IFC	SLD, LST

6. Timeline

IFC / final issues per quarter and discipline (the monthly curve and the AWP bars are in ALP_engineering_timeline.pdf of the engineering_plan engine):

Quarter	CV	CM	EL	GE	HS	HV	IC	ME	PI	PR	PC	SS	Total
2026 Q4	11	1	31	7		6	14	97	6	49		5	227
2027 Q1	59	15	32	82	13	3	73	198	22	10	1		508
2027 Q2	63	42	53	70	12	12	144	130	9	54	1	32	622
2027 Q3	8	43	33	4	11	1	93	108	35	40		5	381
2027 Q4		47	35	18	2		97	89	28				316
2028 Q1		50					4	29	47				130
2028 Q2		14						4	4				22
2028 Q3								1	1				2

Document preparation capacity at full mobilisation (eng_resource, levelled timeline): CV 30 FTE, CM 3 FTE, EL 8 FTE, GE 5 FTE, HS 3 FTE, HV 7 FTE, IC 14 FTE, ME 12 FTE, PI 32 FTE, PR 12 FTE, PC 1 FTE, SS 18 FTE - total 145 FTE.

Documents are scheduled in the order of their latest start (backwards from the EWP and PWP need dates) within these capacities.

Engineering load per month: hours of each EPC document as booked by the levelled timeline (spread over its preparation when not levelled); 174 tender-stage documents (10,413 h) were prepared in the bid and are excluded; FTE at 150 h / month.

Month	CV	CM	EL	GE	HS	HV	IC	ME	PI	PR	PC	SS	Hours	FTE
2026-11	673	170	514	308	172	44	907	758	2,077	762		48	6,434	43
2026-12	3,205	313	919	567	325	731	1,607	1,382	2,970	1,387			13,404	89
2027-01	4,355	426	1,164	718	417	926	1,971	1,736	4,658	1,737			18,107	121
2027-02	4,484	433	1,194	735	435	1,046	2,091	1,628	4,800	1,800	80	1,061	19,787	132
2027-03	5,173	492	1,375	843	511	1,163	2,410	204	5,520	2,065	5	2,018	21,779	145
2027-04	4,946	471	1,319	813	272	523	2,294		5,280	1,977	75	2,709	20,679	138
2027-05	4,186	450	1,259	750	327	36	2,196		5,020	1,878		1,719	17,821	119
2027-06	1,142	484	1,318	224	471	21	2,300		5,243	1,276		124	12,602	84
2027-07	642	477	1,316	53	494		2,303		5,275	9			10,569	70
2027-08	334	474	1,315		364		2,298		5,254				10,040	67
2027-09		466	1,313		75		2,297		5,274				9,426	63
2027-10		446	1,076	36			2,095		5,021				8,674	58
2027-11		473	145	100			997		5,268				6,983	47
2027-12		493		24			147		5,318				5,981	40
2028-01		449							3,609				4,058	27
2028-02		446							1,728				2,174	14
2028-03		302							549				850	6
2028-04		58							634				692	5
2028-05		2							575				577	4
2028-06									250				250	2

Peak 2027-03: 21,779 h (about 145 FTE).

Key dates (CPM, driven by mechanical completion of the CWP's):

Key date	Title	Forecast	Float (wd)
KD-000	Notice to Proceed	2026-11-02	457
KD-010	Back-feed energisation (key date)	2028-07-04	21
KD-020	GT first fire (key date)	2028-09-18	10
KD-030	First synchronisation in combined cycle (key date)	2028-11-20	31
KD-040	Taking-Over of the Works (Time for Completion, 975 days)	2029-04-02	67

7. Supplier documents (VDRL)

750 supplier documents in 66 packages. Per package:

MR	Originator	Docs	Sheets	First IFR	Last final
-	IEC	17	26	2026-09-28	2027-10-04
MR-CV-001	C01	7	7	2027-01-21	2027-02-18
MR-CV-002	C02	7	7	2027-01-21	2027-02-18
MR-CV-003	C03	7	7	2027-02-11	2027-03-11
MR-EL-001	E01	14	71	2027-01-12	2028-04-18
MR-EL-002	E02	14	40	2027-02-11	2028-02-10
MR-EL-003	E03	11	13	2027-02-11	2028-01-13
MR-EL-004	E04	11	18	2027-02-03	2027-10-13
MR-EL-005	E05	12	103	2027-02-11	2027-12-02
MR-EL-006	E06	14	34	2027-02-11	2028-01-13
MR-EL-007	E07	12	87	2027-02-11	2027-10-21
MR-EL-008	E08	3	3	2027-03-25	2027-09-09
MR-EL-009	E09	3	3	2027-02-25	2027-07-15
MR-EL-010	E10	8	8	2027-04-08	2027-10-21
MR-EL-011	E11	8	8	2026-11-02	2027-04-12
MR-EL-012	E12	7	7	2027-03-29	2027-04-26
MR-EL-101	IEC	16	274	2026-09-28	2028-01-31
MR-EL-102	IEC	15	65	2026-09-28	2028-02-21
MR-HV-001	H01	15	356	2027-02-11	2027-12-02
MR-IC-001	I01	12	95	2027-02-18	2028-02-17
MR-IC-002	I02	13	41	2027-02-11	2027-12-02
MR-IC-003	I03	14	56	2027-02-11	2027-12-02
MR-IC-004	I04	3	3	2027-03-11	2027-10-21
MR-IC-005	I05	3	3	2027-03-11	2027-11-18
MR-IC-006	I06	12	86	2027-06-09	2028-01-05
MR-IC-007	I07	12	67	2027-04-29	2027-11-25
MR-IC-008	I08	13	172	2027-04-29	2028-01-06
MR-IC-009	I09	12	26	2027-05-13	2028-04-27
MR-IC-010	I10	3	3	2027-03-11	2027-07-01
MR-IC-101	IEC	19	1584	2026-11-30	2027-11-15
MR-ME-001	M01	15	73	2027-02-11	2028-02-10
MR-ME-002	M02	15	45	2027-02-11	2028-01-13
MR-ME-003	M03	15	45	2027-02-11	2028-03-23
MR-ME-004	M04	15	97	2027-02-11	2028-01-13
MR-ME-005	M05	16	111	2027-02-11	2028-01-13
MR-ME-006	M06	14	46	2027-02-11	2027-12-02
MR-ME-007	M07	15	85	2027-02-11	2027-12-02
MR-ME-008	M08	16	106	2027-02-11	2027-12-02

MR	Originator	Docs	Sheets	First IFR	Last final
MR-ME-009	M09	16	119	2027-02-11	2028-02-10
MR-ME-010	M10	14	55	2027-06-04	2027-12-31
MR-ME-011	M11	14	87	2027-06-01	2028-02-08
MR-ME-012	M12	16	330	2027-02-25	2028-04-06
MR-ME-013	M13	15	149	2027-01-21	2027-09-30
MR-ME-014	M14	16	290	2027-02-11	2027-12-02
MR-ME-015	M15	15	126	2027-02-11	2028-02-10
MR-ME-016	M16	15	152	2027-02-11	2028-01-13
MR-ME-017	M17	7	7	2027-02-18	2027-03-18
MR-ME-018	M18	15	63	2027-02-11	2027-12-02
MR-ME-019	M19	15	38	2027-01-21	2027-09-30
MR-ME-020	M20	15	189	2027-02-11	2027-12-16
MR-ME-021	M21	13	45	2027-02-11	2028-01-13
MR-ME-022	M22	7	7	2027-04-08	2027-05-06
MR-ME-023	M23	7	7	2027-10-12	2027-11-09
MR-ME-024	M24	7	7	2027-09-21	2027-10-19
MR-ME-101	IEC	19	628	2026-09-28	2028-07-24
MR-ME-102	IEC	17	526	2026-09-28	2028-02-07
MR-ME-103	IEC	17	513	2026-09-28	2028-04-03
MR-PI-001	P01	3	3	2027-02-25	2027-08-26
MR-PI-002	P02	3	3	2027-03-11	2028-02-10
MR-PI-003	P03	3	3	2027-03-29	2027-11-08
MR-PI-004	P04	3	3	2027-02-04	2027-10-14
MR-PI-005	P05	3	3	2027-05-05	2027-09-29
MR-PI-006	P06	3	3	2027-02-25	2027-07-22
MR-PI-007	P07	3	3	2027-02-25	2027-08-19
MR-SS-001	S01	23	735	2027-02-11	2027-05-06
MR-SS-002	S02	8	8	2027-02-18	2027-04-01

Power island (IEC) documents:

Number	IEC ref	Title	IFR	IFC	Feeds
ALP-IEC-00000-GE-LST-0001	IEC-ALP-IF-001	Terminal points and interface data - power island	2026-09-28	2026-11-02	1
ALP-IEC-00000-GE-RPT-0001	IEC-ALP-DOR-001	Division of responsibility - power island and balance of plant	2026-09-28	2026-11-02	0
ALP-IEC-00000-GE-SPC-0001	IEC-ALP-REQ-001	Power island interface requirements to the EPC	2026-09-28	2026-11-02	0
ALP-IEC-10BAT-EL-DSH-0001	IEC-ALP-DS-106	Generator step-up transformers datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10BAT-EL-GAD-0001	IEC-ALP-GA-104	Generator step-up transformers - GA, installation and maintenance	2026-09-28	2027-01-11	1
ALP-IEC-10HAA-ME-DSH-0001	IEC-ALP-DS-104	HRSG IE-HR3 datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10HAA-ME-GAD-0001	IEC-ALP-GA-103	HRSG IE-HR3 - GA, erection and maintenance	2026-09-28	2027-01-11	1
ALP-IEC-10MBA-ME-DSH-0001	IEC-ALP-DS-101	Gas turbine IE-9H.02 datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10MBA-ME-GAD-0001	IEC-ALP-GA-101	Gas turbine and generator - GA, maintenance and installation	2026-09-28	2027-01-11	1
ALP-IEC-10MBA-ME-LAY-0001	IEC-ALP-GA-100	Power island arrangement - installation and maintenance spaces	2026-09-28	2027-01-11	1
ALP-IEC-10MBA-PR-HMB-0001	IEC-ALP-PER-001	Power island thermal performance data (heat balance)	2026-09-28	2026-11-02	1
ALP-IEC-10MBA-PR-HMB-0002	IEC-ALP-PER-003	Power island off-design heat balances (case book)	2026-09-28	2026-11-02	0
ALP-IEC-10MKA-EL-DSH-0001	IEC-ALP-DS-102	Generators, GCB and isolated phase busduct datasheet	2026-09-28	2026-12-28	1
ALP-IEC-20MAA-ME-DSH-0001	IEC-ALP-DS-103	Steam turbine IE-ST3R datasheet	2026-09-28	2026-12-28	0
ALP-IEC-20MAA-ME-GAD-0001	IEC-ALP-GA-102	Steam turbine, generator and condenser - GA, maintenance	2026-09-28	2027-01-11	1
ALP-IEC-20MAG-ME-DSH-0001	IEC-ALP-DS-105	Seawater-cooled surface condenser datasheet	2026-09-28	2026-12-28	0
ALP-IEC-10BAT-ME-ITP-0001	-	Supplier inspection and test plan - Generator step-up transformers (IEC)	2026-11-30	2026-12-28	0
ALP-IEC-10HAA-ME-ITP-0001	-	Supplier inspection and test plan - HRSG incl. SCR (IEC)	2026-11-30	2026-12-28	0
ALP-IEC-10MBY-ME-DSH-0001	-	Supplier certified datasheets and performance curves - Turbine control system (IEC)	2026-11-30	2027-01-11	0
ALP-IEC-10MBY-ME-ITP-0001	-	Supplier inspection and test plan - Turbine control system (IEC)	2026-11-30	2026-12-28	0
ALP-IEC-10MKA-ME-ITP-0001	-	Supplier inspection and test plan - Generators, excitation, GCB and busducts (IEC)	2026-11-30	2026-12-28	0
ALP-IEC-20MAA-ME-ITP-0001	-	Supplier inspection and test plan - Steam turbine and condenser (IEC)	2026-11-30	2026-12-28	0
ALP-IEC-10BAT-CV-CAL-0001	-	Supplier foundation loads and anchor bolt drawings - Generator step-up transformers (IEC)	2026-12-14	2027-01-11	1
ALP-IEC-10HAA-CV-CAL-0001	IEC-ALP-FL-102	Foundation load data and anchor bolt drawings - HRSG	2026-12-14	2027-01-11	13
ALP-IEC-10MBA-CV-CAL-0001	IEC-ALP-FL-101	Foundation load data and anchor bolt drawings - GT and generator	2026-12-14	2027-01-11	14
ALP-IEC-10MBY-ME-GAD-0001	-	Supplier GA drawings - Turbine control system (IEC)	2026-12-14	2027-01-25	0
ALP-IEC-10MKA-CV-CAL-0001	-	Supplier foundation loads and anchor bolt drawings - Generators, excitation, GCB and busducts (IEC)	2026-12-14	2027-01-11	2
ALP-IEC-10MKA-ME-GAD-0001	-	Supplier GA drawings - Generators, excitation, GCB and busducts (IEC)	2026-12-14	2027-01-25	0
ALP-IEC-00000-PI-LST-0001	IEC-ALP-IF-001-A	Terminal point nozzle loads (IF-001 attachment A)	2026-12-28	2027-01-25	9
ALP-IEC-10BAA-EL-SLD-0001	IEC-ALP-SLD-101	Power island single line diagram (generators, GCB, GSU, UAT tap-offs, SFC)	2026-12-28	2027-02-08	1
ALP-IEC-10BAT-IC-IOL-0001	-	Supplier I/O list, control philosophy and interface signals - Generator step-up transformers (IEC)	2026-12-28	2027-02-08	0

Number	IEC ref	Title	IFR	IFC	Feeds
ALP-IEC-10BAT-ME-CAL-0001	-	Supplier design calculations - Generator step-up transformers (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10HAA-IC-IOL-0001	-	Supplier I/O list, control philosophy and interface signals - HRSG incl. SCR (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10HAA-ME-CAL-0001	-	Supplier design calculations - HRSG incl. SCR (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10MBA-EL-LST-0001	IEC-ALP-EL-101	Power island electrical load list	2026-12-28	2027-02-08	1
ALP-IEC-10MBA-IC-IOL-0001	-	Supplier I/O list, control philosophy and interface signals - Gas turbine package incl. auxiliaries (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10MBA-ME-CAL-0001	-	Supplier design calculations - Gas turbine package incl. auxiliaries (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10MBY-ME-CAL-0001	-	Supplier design calculations - Turbine control system (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10MKA-IC-IOL-0001	-	Supplier I/O list, control philosophy and interface signals - Generators, excitation, GCB and busducts (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10MKA-ME-CAL-0001	-	Supplier design calculations - Generators, excitation, GCB and busducts (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-20MAA-CV-CAL-0001	IEC-ALP-FL-103	Foundation load data - ST, generator and condenser	2026-12-28	2027-01-25	5
ALP-IEC-20MAA-IC-IOL-0001	-	Supplier I/O list, control philosophy and interface signals - Steam turbine and condenser (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-20MAA-ME-CAL-0001	-	Supplier design calculations - Steam turbine and condenser (IEC)	2026-12-28	2027-02-08	0
ALP-IEC-10BAT-EL-WIR-0001	-	Supplier electrical schematics and wiring diagrams - Generator step-up transformers (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-10BAT-IC-DSH-0001	-	Supplier instrument list and datasheets - Generator step-up transformers (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-10HAA-EL-WIR-0001	-	Supplier electrical schematics and wiring diagrams - HRSG incl. SCR (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-10HAA-IC-DSH-0001	-	Supplier instrument list and datasheets - HRSG incl. SCR (IEC) - part 1 of 2	2027-01-11	2027-03-08	0
ALP-IEC-10HAA-IC-DSH-0002	-	Supplier instrument list and datasheets - HRSG incl. SCR (IEC) - part 2 of 2	2027-01-11	2027-03-08	0
ALP-IEC-10HAA-PR-PID-0001	IEC-ALP-PID-102	HRSG P&IDs incl. SCR and ammonia injection	2027-01-11	2027-03-08	2
ALP-IEC-10MBA-EL-WIR-0001	-	Supplier electrical schematics and wiring diagrams - Gas turbine package incl. auxiliaries (IEC) - part 1 of 2	2027-01-11	2027-03-08	0
ALP-IEC-10MBA-EL-WIR-0002	-	Supplier electrical schematics and wiring diagrams - Gas turbine package incl. auxiliaries (IEC) - part 2 of 2	2027-01-11	2027-03-08	0
ALP-IEC-10MBA-HS-RPT-0001	IEC-ALP-HS-101	Power island noise data (ISO 10494)	2027-01-11	2027-02-22	1
ALP-IEC-10MBA-IC-DSH-0001	-	Supplier instrument list and datasheets - Gas turbine package incl. auxiliaries (IEC) - part 1 of 3	2027-01-11	2027-03-08	0
ALP-IEC-10MBA-IC-DSH-0002	-	Supplier instrument list and datasheets - Gas turbine package incl. auxiliaries (IEC) - part 2 of 3	2027-01-11	2027-03-08	0
ALP-IEC-10MBA-IC-DSH-0003	-	Supplier instrument list and datasheets - Gas turbine package incl. auxiliaries (IEC) - part 3 of 3	2027-01-11	2027-03-08	0
ALP-IEC-10MBA-PR-PID-0001	IEC-ALP-PID-101	Gas turbine P&IDs (fuel gas, liquid fuel, lube oil, inlet air, washing)	2027-01-11	2027-03-08	2
ALP-IEC-10MBY-EL-WIR-0001	-	Supplier electrical schematics and wiring diagrams - Turbine control system (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-10MBY-IC-BLK-0001	IEC-ALP-IC-101	TCS architecture and DCS integration (OPC UA, IEC 62443 zone)	2027-01-11	2027-02-22	1
ALP-IEC-10MKA-EL-WIR-0001	-	Supplier electrical schematics and wiring diagrams - Generators, excitation, GCB and busducts (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-10MKA-IC-DSH-0001	-	Supplier instrument list and datasheets - Generators, excitation, GCB and busducts (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-20MAA-EL-WIR-0001	-	Supplier electrical schematics and wiring diagrams - Steam turbine and condenser (IEC)	2027-01-11	2027-03-08	0
ALP-IEC-20MAA-IC-DSH-0001	-	Supplier instrument list and datasheets - Steam turbine and condenser (IEC) - part 1 of 2	2027-01-11	2027-03-08	0
ALP-IEC-20MAA-IC-DSH-0002	-	Supplier instrument list and datasheets - Steam turbine and condenser (IEC) - part 2 of 2	2027-01-11	2027-03-08	0
ALP-IEC-10BAT-ME-LST-0001	-	Spare parts and special tools lists - Generator step-up transformers (IEC)	2027-01-25	2027-03-22	0
ALP-IEC-10HAA-ME-LST-0001	-	Spare parts and special tools lists - HRSG incl. SCR (IEC)	2027-01-25	2027-03-22	0
ALP-IEC-10MBA-ME-ITP-0001	IEC-ALP-QA-101	Power island inspection and test plans	2027-01-25	2027-02-22	0
ALP-IEC-10MBA-ME-LST-0001	-	Spare parts and special tools lists - Gas turbine package incl. auxiliaries (IEC)	2027-01-25	2027-03-22	0
ALP-IEC-10MBY-ME-LST-0001	-	Spare parts and special tools lists - Turbine control system (IEC)	2027-01-25	2027-03-22	0
ALP-IEC-10MKA-EL-RPT-0001	IEC-ALP-EL-102	Generator capability curves and dynamic models (Grid Code)	2027-01-25	2027-03-22	1
ALP-IEC-10MKA-ME-LST-0001	-	Spare parts and special tools lists - Generators, excitation, GCB and busducts (IEC)	2027-01-25	2027-03-22	0
ALP-IEC-10MKA-PR-PID-0001	IEC-ALP-PID-104	Generator auxiliary P&IDs (H2/CO2, seal oil, stator water)	2027-01-25	2027-03-22	0
ALP-IEC-20MAA-ME-LST-0001	-	Spare parts and special tools lists - Steam turbine and condenser (IEC)	2027-01-25	2027-03-22	0
ALP-IEC-20MAA-PR-PID-0001	IEC-ALP-PID-103	Steam turbine and condenser P&IDs (lube, gland, vacuum, EHC, bypass valves)	2027-01-25	2027-03-22	1
ALP-IEC-10MBY-IC-CNR-0001	IEC-ALP-IC-103	Power island operating and control description (start-up, load, trips)	2027-02-08	2027-04-05	2
ALP-IEC-10MBA-PI-M3D-0001	IEC-ALP-M3D-101	Power island 3D model export (for the plant model)	2027-02-22	2027-04-19	0

Number	IEC ref	Title	IFR	IFC	Feeds
ALP-IEC-10MBY-IC-IOL-0001	IEC-ALP-IC-102	TCS I/O list and interface signal list	2027-02-22	2027-04-19	3
ALP-IEC-10MBY-IC-LOG-0001	-	Supplier logic diagrams - Turbine control system (IEC) - part 1 of 6	2027-03-22	2027-06-14	0
ALP-IEC-10MBY-IC-LOG-0002	-	Supplier logic diagrams - Turbine control system (IEC) - part 2 of 6	2027-03-22	2027-06-14	0
ALP-IEC-10MBY-IC-LOG-0003	-	Supplier logic diagrams - Turbine control system (IEC) - part 3 of 6	2027-03-22	2027-06-14	0
ALP-IEC-10MBY-IC-LOG-0004	-	Supplier logic diagrams - Turbine control system (IEC) - part 4 of 6	2027-03-22	2027-06-14	0
ALP-IEC-10MBY-IC-LOG-0005	-	Supplier logic diagrams - Turbine control system (IEC) - part 5 of 6	2027-03-22	2027-06-14	0
ALP-IEC-10MBY-IC-LOG-0006	-	Supplier logic diagrams - Turbine control system (IEC) - part 6 of 6	2027-03-22	2027-06-14	0
ALP-IEC-10MBY-CM-CMP-0001	-	Supplier commissioning procedures - Turbine control system (IEC)	2027-04-05	2027-06-28	0
ALP-IEC-10MBY-ME-TST-0001	-	Factory acceptance test procedure - Turbine control system (IEC)	2027-05-03	2027-06-14	0
ALP-IEC-10MBY-IC-CBL-0001	IEC-ALP-IC-104	Power island cable list (IEC scope and interfaces)	2027-05-31	2027-07-26	2
ALP-IEC-10MBY-ME-PRC-0001	-	Supplier installation and erection manual - Turbine control system (IEC)	2027-05-31	2027-07-12	0
ALP-IEC-10MKA-CM-CMP-0001	-	Supplier commissioning procedures - Generators, excitation, GCB and busducts (IEC)	2027-06-21	2027-09-13	0
ALP-IEC-10HAA-CM-CMP-0001	-	Supplier commissioning procedures - HRSG incl. SCR (IEC)	2027-06-28	2027-09-20	0
ALP-IEC-10MBA-CM-CMP-0001	-	Supplier commissioning procedures - Gas turbine package incl. auxiliaries (IEC)	2027-06-28	2027-09-20	0
ALP-IEC-10MBY-ME-OMM-0001	-	Supplier operation and maintenance manual - Turbine control system (IEC)	2027-06-28	2027-11-15	0
ALP-IEC-10BAT-CM-CMP-0001	-	Supplier commissioning procedures - Generator step-up transformers (IEC)	2027-07-12	2027-10-04	0
ALP-IEC-10MBA-ME-PRC-0001	IEC-ALP-ER-101	Power island erection manuals and procedures	2027-07-12	2027-09-06	0
ALP-IEC-10MKA-ME-TST-0001	-	Factory acceptance test procedure - Generators, excitation, GCB and busducts (IEC)	2027-07-19	2027-08-30	0
ALP-IEC-10HAA-ME-TST-0001	-	Factory acceptance test procedure - HRSG incl. SCR (IEC)	2027-07-26	2027-09-06	0
ALP-IEC-10MBA-ME-TST-0001	-	Factory acceptance test procedure - Gas turbine package incl. auxiliaries (IEC)	2027-07-26	2027-09-06	0
ALP-IEC-10MBY-ME-TRP-0001	-	Factory acceptance test report - Turbine control system (IEC)	2027-07-26	2027-08-09	0
ALP-IEC-10BAT-ME-TST-0001	-	Factory acceptance test procedure - Generator step-up transformers (IEC)	2027-08-09	2027-09-20	0
ALP-IEC-10MBA-ME-RPT-0001	IEC-ALP-OM-102	Maintenance concept and long-term service offer	2027-08-09	2027-10-04	0
ALP-IEC-10MBY-ME-MDB-0001	-	Manufacturing record book and conformity certificates - Turbine control system (IEC)	2027-08-09	2027-10-04	0
ALP-IEC-10MKA-ME-PRC-0001	-	Supplier installation and erection manual - Generators, excitation, GCB and busducts (IEC)	2027-08-16	2027-09-27	0
ALP-IEC-10HAA-ME-PRC-0001	-	Supplier installation and erection manual - HRSG incl. SCR (IEC)	2027-08-23	2027-10-04	0
ALP-IEC-20MAA-CM-CMP-0001	-	Supplier commissioning procedures - Steam turbine and condenser (IEC)	2027-08-23	2027-11-15	0
ALP-IEC-10BAT-ME-PRC-0001	-	Supplier installation and erection manual - Generator step-up transformers (IEC)	2027-09-06	2027-10-18	0
ALP-IEC-10MKA-ME-OMM-0001	-	Supplier operation and maintenance manual - Generators, excitation, GCB and busducts (IEC)	2027-09-13	2028-01-31	0
ALP-IEC-10HAA-ME-OMM-0001	-	Supplier operation and maintenance manual - HRSG incl. SCR (IEC)	2027-09-20	2028-02-07	0
ALP-IEC-20MAA-ME-TST-0001	-	Factory acceptance test procedure - Steam turbine and condenser (IEC)	2027-09-20	2027-11-01	0
ALP-IEC-10BAT-ME-OMM-0001	-	Supplier operation and maintenance manual - Generator step-up transformers (IEC)	2027-10-04	2028-02-21	0
ALP-IEC-10MKA-ME-TRP-0001	-	Factory acceptance test report - Generators, excitation, GCB and busducts (IEC)	2027-10-11	2027-10-25	0
ALP-IEC-10HAA-ME-TRP-0001	-	Factory acceptance test report - HRSG incl. SCR (IEC)	2027-10-18	2027-11-01	0
ALP-IEC-10MBA-ME-TRP-0001	-	Factory acceptance test report - Gas turbine package incl. auxiliaries (IEC)	2027-10-18	2027-11-01	0
ALP-IEC-20MAA-ME-PRC-0001	-	Supplier installation and erection manual - Steam turbine and condenser (IEC)	2027-10-18	2027-11-29	0
ALP-IEC-10MKA-ME-MDB-0001	-	Manufacturing record book and conformity certificates - Generators, excitation, GCB and busducts (IEC)	2027-10-25	2027-12-20	0
ALP-IEC-10BAT-ME-TRP-0001	-	Factory acceptance test report - Generator step-up transformers (IEC)	2027-11-01	2027-11-15	0
ALP-IEC-10HAA-ME-MDB-0001	-	Manufacturing record book and conformity certificates - HRSG incl. SCR (IEC)	2027-11-01	2027-12-27	0
ALP-IEC-10MBA-ME-MDB-0001	-	Manufacturing record book and conformity certificates - Gas turbine package incl. auxiliaries (IEC)	2027-11-01	2027-12-27	0
ALP-IEC-10BAT-ME-MDB-0001	-	Manufacturing record book and conformity certificates - Generator step-up transformers (IEC)	2027-11-15	2028-01-10	0
ALP-IEC-20MAA-ME-OMM-0001	-	Supplier operation and maintenance manual - Steam turbine and condenser (IEC)	2027-11-15	2028-04-03	0
ALP-IEC-20MAA-ME-TRP-0001	-	Factory acceptance test report - Steam turbine and condenser (IEC)	2027-12-13	2027-12-27	0
ALP-IEC-20MAA-ME-MDB-0001	-	Manufacturing record book and conformity certificates - Steam turbine and condenser (IEC)	2027-12-27	2028-02-21	0
ALP-IEC-10MBA-ME-OMM-0001	IEC-ALP-OM-101	Power island operation and maintenance manuals	2028-03-06	2028-07-24	0

8. Near-critical items

EWPs with less than 20 working days float (the document with the smallest float drives the EWP):

EWP	CWP	Driving document	Ready	Needed	Float
EWP-11-CV-01	CWP-11-CV-01	ALP-EPC-00UCA-HV-GAD-0002	2027-04-26	2027-03-29	4
EWP-06-CV-01	CWP-06-CV-01	ALP-EPC-00UPA-EL-ELY-0001	2027-09-29	2027-09-03	8
EWP-12-PI-01	CWP-12-PI-01	ALP-EPC-00SGA-PI-ISO-0001	2027-10-20	2027-07-12	8
EWP-12-CV-01	CWP-12-CV-01	ALP-EPC-00UZA-PI-LAY-0001	2027-01-11	2027-01-18	9
EWP-08-CV-01	CWP-08-CV-01	ALP-EPC-00UQA-HV-GAD-0001	2027-06-03	2027-05-04	10
EWP-03-CV-01	CWP-03-CV-01	ALP-EPC-00UBA-CV-RBR-0001	2027-04-23	2027-03-29	11
EWP-01-CV-02	CWP-01-CV-02	ALP-EPC-10UHA-CV-RBR-0001	2027-06-18	2027-04-19	12
EWP-02-PI-01	CWP-02-PI-01	ALP-EPC-20LCA-PI-SUP-0001	2028-04-27	2028-01-31	12
EWP-05-CV-01	CWP-05-CV-01	ALP-EPC-00UPC-CV-RBR-0001	2027-06-03	2027-03-25	14
EWP-10-CV-01	CWP-10-CV-01	ALP-EPC-00UEG-CV-RBR-0001	2027-06-28	2027-06-03	14
EWP-04-CV-01	CWP-04-CV-01	ALP-EPC-00UAC-CV-RBR-0001	2027-04-16	2027-04-05	15
EWP-07-CV-01	CWP-07-CV-01	ALP-EPC-00UGA-CV-EFD-0001	2027-05-31	2027-04-05	15
EWP-07-PI-01	CWP-07-PI-01	ALP-EPC-00GHA-PI-SUP-0001	2028-03-13	2028-01-24	15
EWP-01-PI-01	CWP-01-PI-01	ALP-EPC-10LAF-PI-SUP-0001	2028-03-02	2027-12-13	17
EWP-02-CV-02	CWP-02-CV-02	ALP-EPC-20UMB-CV-RBR-0001	2027-06-28	2027-04-19	18

PWPs (MRs) with less than 20 working days float between delivery on site and the first CWP start:

MR	Title	PO	On site	Needed	CWP	Float
MR-ME-101	Gas turbine package incl. auxiliaries (IEC)	2026-11-02	2027-11-15	2027-11-15	CWP-01-ME-01	0
MR-ME-102	HRSR incl. SCR (IEC)	2026-11-02	2027-11-15	2027-11-15	CWP-01-ME-01	0
MR-EL-001	380 kV GIS	2026-12-15	2028-01-25	2028-02-01	CWP-04-EL-01	5
MR-EL-011	Construction power supply (RMU, temporary distribution)	2026-11-16	2027-01-18	2027-01-25	CWP-12-CV-01	5
MR-EL-101	Generators, excitation, GCB and busducts (IEC)	2026-11-02	2027-11-08	2027-11-15	CWP-01-ME-01	5
MR-ME-103	Steam turbine and condenser (IEC)	2026-11-02	2028-01-10	2028-01-17	CWP-02-ME-01	5
MR-EL-009	Cable trays, earthing and lighting materials	2027-01-28	2027-06-03	2027-06-14	CWP-12-EL-01	7
MR-ME-001	Boiler feed pumps	2027-01-14	2027-11-18	2027-11-29	CWP-08-ME-01	7
MR-ME-009	Auxiliary boiler	2027-01-14	2027-11-18	2027-11-29	CWP-08-ME-01	7
MR-ME-012	Water treatment package (SWRO, demin, condensate polishing)	2027-01-28	2028-01-13	2028-01-24	CWP-07-ME-01	7
MR-ME-015	Gas receiving and metering station	2027-01-14	2027-11-18	2027-11-29	CWP-09-ME-01	7
MR-ME-016	LDO unloading and forwarding / treatment skids	2027-01-14	2027-10-21	2027-11-01	CWP-10-ME-01	7
MR-PI-002	Alloy piping P91 / P22 (main steam, reheat)	2027-02-11	2027-12-30	2028-01-10	CWP-01-PI-01	7
MR-ME-011	Chemical dosing skid	2027-05-04	2027-11-16	2027-11-29	CWP-08-ME-01	9
MR-CV-002	Civil and building works	2026-12-24	2027-01-07	2027-01-25	CWP-12-CV-01	12
MR-ME-003	CW pumps	2027-01-14	2027-12-30	2028-01-17	CWP-05-ME-01	12
MR-IC-001	Plant DCS	2027-01-21	2027-11-25	2027-12-20	CWP-11-IC-01	17
MR-ME-021	EOT cranes	2027-01-14	2027-10-21	2027-11-15	CWP-01-ME-01	17
MR-PI-001	Piping bulk materials (carbon steel, stainless)	2027-01-28	2027-07-15	2027-08-09	CWP-12-PI-01	17

9. Limitations of this revision

- Dates are levelled against the document preparation capacity of section 6 (basis decision:DEC-EPCE-0010); a change of staffing changes the dates at the next run.
- Document quantities are estimates per system until the P&IDs and the 3D model produce the instrument, line and valve registers; the rules then use the register counts (section 3.1).
- Benchmarks are indicative ranges (engineering judgement), to be replaced by company historical data.
- Supplier document quantities follow the package rules; the supplier's own VDRL replaces them after the PO.
- KKS key titles are not yet verified against the licensed VGB key catalogue (Owner's Engineer, TQ-035).

Check results (engine)

29 OK, 6 INFO, 0 WARN (checks: MDL rules, benchmark, coverage MR, coverage equipment, coverage requirement, coverage scope, coverage system, design criteria, inputs, key date). WARN must be resolved before the next issue; INFO is noted.

Item	Check	Result	Detail
-	MDL rules	OK	80 rules: 0 required document(s) missing, 0 to update, 0 orphan(s)
ALP-EPC-00000-CM-OMM-0002	inputs	INFO	none of the typical inputs ['CMP', 'PID'] of a OMM
ALP-EPC-00000-EL-LST-0002	inputs	INFO	none of the typical inputs ['DBR', 'HAZ', 'LST', 'PFD', 'PID', 'PRC', 'RPT', 'SLD', 'STU'] of a LST
ALP-EPC-00000-PI-M3D-0004	inputs	INFO	none of the typical inputs ['ISO', 'LAY', 'PLT', 'PPL'] of a M3D

Item	Check	Result	Detail
ALP-EPC-00UPA-CV-RBR-0001	inputs	INFO	none of the typical inputs ['FND'] of a RBR
ALP-EPC-00UZC-CV-RBR-0001	inputs	INFO	none of the typical inputs ['FND'] of a RBR
ALP-EPC-00UZE-CV-RBR-0001	inputs	INFO	none of the typical inputs ['FND'] of a RBR
-	coverage requirement	OK	150 of 150 requirements answered
-	coverage scope	OK	88 of 88 EPC/IEC scope items covered